

I3L: Iterative Interactive Imitation Learning for Whole-Body Mobile Manipulation

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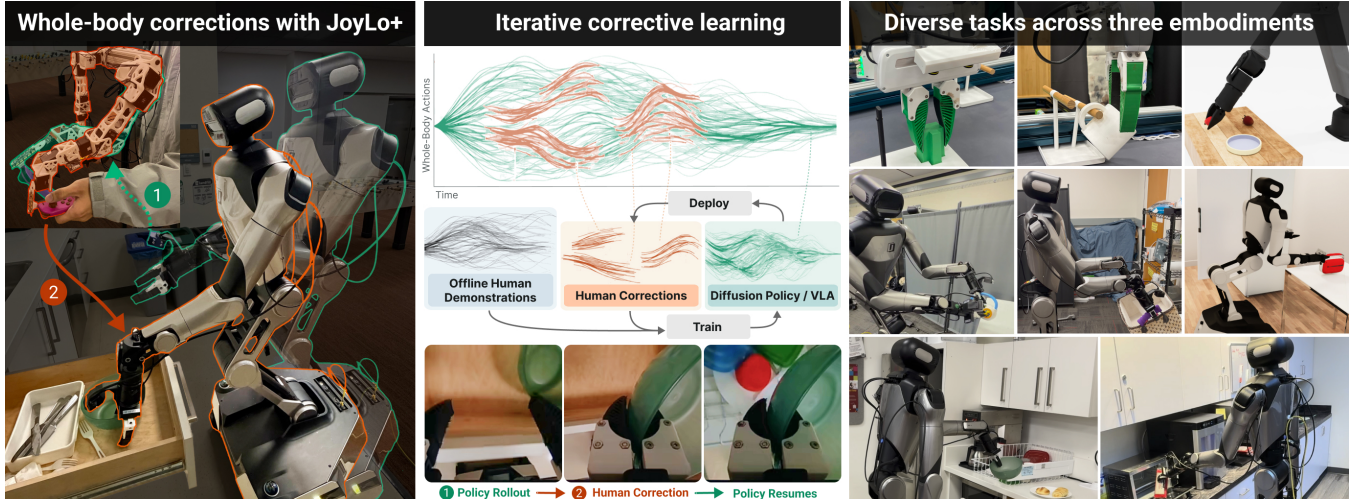


Fig. 1. I3L couples interactive correction with iterative policy refinement. Left: 1) the policy approaches a failure, prompting the operator to take over with JoyLo+ and provide targeted corrections, 2) the operator uses JoyLo+ to take over during autonomous execution. Center: corrections are aggregated with prior demonstrations for policy retraining and redeployment. Right: representative tasks and embodiments.

Abstract—Policies trained offline from teleoperated demonstrations can degrade outside their training distribution. This gap is particularly acute in long-horizon whole-body mobile manipulation, where long horizons amplify compounding errors and coordinated arm, torso, and base motion makes exhaustive demonstration coverage impractical. We introduce I3L, an integrated hardware and learning system that couples targeted whole-body correction with iterative policy refinement to address this last-mile reliability gap. Its low-cost, open-source JoyLo+ interface uses policy-tracking leader arms and touch-triggered takeover to enable coordinated corrections across the body parts, while repeated deployment and retraining allow operators to target failures of the updated policy. We evaluate I3L across three robot embodiments and eight simulated and real-world tasks, including four long-horizon whole-body mobile manipulation tasks. Controlled simulation comparisons show that four-round HG-Dagger achieves the highest final success on both simulation tasks. Under this protocol, I3L improves autonomous success over the base policies by an average of 68.3 percentage points and achieves at least 90% success on every task, outperforming demonstration-only training by 35.6 percentage points on average under matched training-trajectory budgets. Across four rounds, task-averaged takeover duration per episode decreases by 72%. Together, these results show that I3L helps close the last-mile gap between offline training and reliable deployment and provides practical insights into collecting and using corrections for long-horizon whole-body mobile manipulation. Videos, visualizations, and appendix are at <https://i3l-icra.github.io/>.

I. INTRODUCTION

Visuomotor imitation learning has advanced through improved action representations and policy learning methods [1, 2], alongside generalist robot policies trained on

increasingly diverse datasets [3–9]. These developments enable language-conditioned manipulation, cross-embodiment transfer, and long-horizon mobile manipulation. Yet when one is deploying a robot to finish a real task, it still fails in long, frustrating ways. Many policies use *offline imitation learning* from fixed demonstrations, leaving them vulnerable to distribution shift and compounding errors during deployment. This covariate-shift problem motivates expert supervision on policy-visited states in DAgger and noise-injected demonstrations in DART [10, 11].

This challenge is prevalent in whole-body mobile manipulation, which couples over twenty actuated degrees of freedom with an omnidirectional base across varied scenes and objects. In long-horizon tasks, a policy may perform most stages competently yet remain unreliable at full-task completion because of critical failure-prone bottlenecks. Closing this *last-mile gap* therefore depends not just on collecting more data, but on where supervision is provided: targeted corrections at bottleneck states can be more valuable than demonstrations of already-mastered behavior.

Interactive imitation learning (IIL) lets a human supervise the policy at deployment and correct it where it fails, so the next iteration trains on the policy’s own induced distribution [10, 12–17]. Targeted intervention concentrates teleoperation at failure-prone states, reducing the need for end-to-end human control. Prior work has demonstrated repeated corrective learning, compliant intervention interfaces, and mobile bimanual manipulation [14, 16, 17].

Despite this progress, systematic evidence remains limited

for corrective learning across diverse long-horizon tasks requiring coordinated base, torso, and arm motion. Making such interventions practical requires an interface that lets the operator take control promptly while remaining aligned with autonomous execution. Effective corrective learning also requires choosing how to train on corrections and whether to collect them against a fixed or updated policy.

To this end, we present **I3L** (Iterative Interactive Imitation Learning), an integrated hardware and learning system for iterative whole-body correction in long-horizon mobile manipulation. Its JoyLo+ interface combines policy-tracking leader arms with capacitive-touch-triggered switching to support smooth takeover, while repeated deployment and retraining allow collection to target failures of the updated policy. Using this system, we evaluate iterative whole-body correction on four long-horizon real-world BEHAVIOR tasks [18], complemented by controlled comparisons of collection schedules and learning strategies and analysis of intervention burden across simulation and real-world tasks. Our contributions are:

1. An integrated system for iterative whole-body correction. I3L combines JoyLo+ with repeated policy deployment and retraining for long-horizon mobile manipulation. On all four real-world BEHAVIOR tasks [18], I3L helps close the last-mile reliability gap by achieving over 90% success, outperforming demonstration-only training by 30 percentage points on average under matched trajectory budgets.

2. JoyLo+: a touch-triggered whole-body correction interface. Extending GELLO [19] and JoyLo [20], our low-cost, open-source interface combines capacitive-touch detection with position/current-mode switching on motorized leader arms, enabling global takeover for coordinated arm, torso, and base corrections.

3. Studies of corrective learning and operator intervention. Under matched trajectory budgets, we compare single-round and multi-round collection across four correction-learning methods on two simulation tasks. The results reveal strategy-dependent benefits of iteration, with four-round HG-Dagger [12] achieving the highest final mean success on both simulation tasks. A user study also shows declining cumulative intervention share across operators with varied experience.

II. RELATED WORK

Data-collection Interfaces for Correction. Teleoperation interfaces shape the fidelity and ease of collecting corrections. Existing approaches use 3D mice [15], smartphones [13], and motion capture [21]. TeleMoMa [22] combines heterogeneous inputs for whole-body mobile teleoperation, while ALOHA, Mobile ALOHA, GELLO, and JoyLo use kinematic-twin interfaces [2, 19, 20, 23]. RoboCopilot and HACTS synchronize their leaders with the robot during autonomous execution to support takeover [16, 24]; CR-Dagger supports kinesthetic corrections through a force-instrumented handle [17]. Human-Robot Copilot uses motion scaling for fine-grained corrective teleoperation [25], and DexHiL integrates coordinated arm-hand intervention for

dexterous VLA post-training [26]. I3L’s JoyLo+ combines policy-tracking arm leaders with touch-triggered takeover for coordinated arm, torso, and base corrections.

Interactive Imitation Learning Algorithms. DAgger addresses covariate shift by collecting expert labels on policy-visited states [10]. HG-Dagger lets humans gate intervention [12], an approach also used in BC-Z [27]; SafeDagger reduces expert queries with a learned safety classifier [28], while ThriftyDagger gates queries using novelty and risk estimates [29]. Methods also differ in how they use deployment data: IWR balances intervention and non-intervention samples [13], SIRIUS reweights data using estimated human trust [14], and DexHiL prioritizes corrective segments [26]. TRANSIC and CR-Dagger learn residual corrections [15, 17], while FlowCorrect adapts flow policies from sparse relative corrections with a frozen backbone [30]. RECAP combines demonstrations, autonomous experience, and interventions through advantage-conditioned reinforcement learning [31]. I3L compares correction-learning strategies and single-round versus multi-round collection under matched trajectory budgets.

Tasks and Domains. Much interactive-IL evaluation focuses on tabletop manipulation [13–15, 17]. Whole-body and mobile policy learning has advanced through Mobile ALOHA, HumanPlus, BRS, and generalist VLAs [8, 9, 20, 23, 32]. RoboCopilot evaluates corrective learning on mobile part transport and a separate four-stage kitchen task [16], while XR-2 reports iterative DAgger results for clothes folding [33]. Crucially, these studies leave open how correction collection schedules and learning strategies affect performance and intervention burden across diverse long-horizon whole-body mobile-manipulation tasks. I3L addresses this gap through four mobile household tasks with task-stage and body-part intervention analysis, complemented by matched-budget simulation comparisons.

III. METHOD

I3L integrates JoyLo+ correction hardware with iterative policy learning. JoyLo+ lets operators follow the policy’s motion, take over at failure states, and guide coordinated recovery. The system records these corrections with intervention labels, aggregates them with prior demonstrations, and updates the policy for the next deployment round.

A. JoyLo+: A Hardware Interface for Policy Intervention

JoyLo+ is an actuated interface for intervention during autonomous policy execution. We implement **R1 Pro JoyLo+**, with two seven-joint leaders for bimanual whole-body mobile manipulation, and **Franka JoyLo+**, with one seven-joint leader for single-arm table-top manipulation. Building on GELLO’s kinematic leaders [19] and JoyLo’s interface [20], both combine policy-target tracking, capacitive-touch detection, and compliant hand guidance (Figure 2). Table I compares their capabilities with existing interfaces.

a) *Actuated leaders for tracking and correction:* Each configuration uses 3D-printed links, joint mappings, and calibration matched to its robot. During policy rollouts,

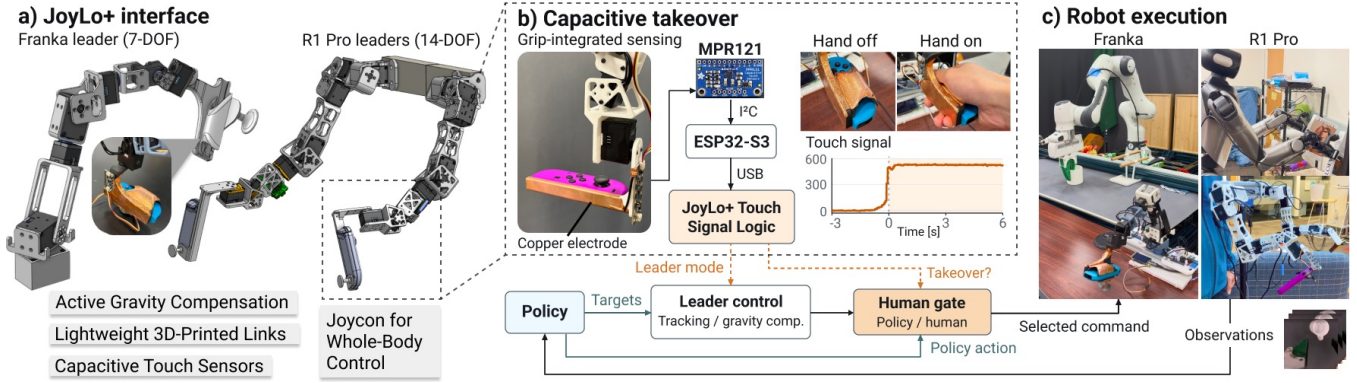


Fig. 2. **JoyLo+ correction interface.** Motorized arm leaders track policy targets during autonomous execution and switch to compliant current control during global takeover. Capacitive touch sensors trigger takeover, while JoyCon inputs control the torso, mobile base, and grippers. The interface supports single-arm and dual-arm configurations.

TABLE I: Teleoperation and correction interfaces. ✓: demonstrated; –: not demonstrated in the cited work.

System	Mobile	Bimanual	Multi-embodiment	Actuated	Correction
ALOHA/2 [2, 34]	–	✓	–	–	–
GELLO [19]	–	✓	✓	–	–
Mobile ALOHA [23]	✓	✓	–	–	–
TeleMoMa [22]	✓	✓	✓	–	–
JoyLo [20]	✓	✓	–	✓	–
UME [35]	✓	✓	✓	✓	–
CR-Dagger [17]	–	–	–	–	✓
HACTS [24]	–	–	–	✓	✓
RoboCopilot [16]	✓	✓	–	✓	✓
JoyLo+ (ours)	✓	✓	✓	✓	✓

Mobile: coordinated arm and mobile-base control; torso where available. *Multi-embodiment:* distinct arm kinematics on real hardware. *Actuated:* leader motors respond to robot state, torque, or policy targets. *Correction:* takeover or kinesthetic guidance during autonomous rollouts. JoyLo+: R1 Pro (mobile, bimanual); R1 Pro and Franka (multi-embodiment).

the actuated leaders track the policy’s arm joint targets, physically displaying the commanded configuration on the device used for correction. During intervention, gravity compensation supports the leaders while the operator guides their motion. The component cost for one R1 Pro leader assembly and the listed accessories totals \$1,257.49. A detailed cost breakdown is available on our project website.

b) Grip-integrated sensing for direct takeover: Both configurations use grip-integrated copper electrodes, an MPR121 capacitive-sensing board, and an ESP32-S3 controller to report touch state to the host. Grasping the interface requests takeover without first displacing a tracking leader. Separate touch and release thresholds and a release cooldown maintain takeover through brief interruptions in contact.

c) Single-arm and whole-body intervention: On R1 Pro, touching a grip or providing a JoyCon (with joysticks and buttons) input transfers command of both arms, the torso, the base, and the grippers together. The leaders guide the arms, while JoyCons control the torso, base, and grippers. In both configurations, releasing the interface returns control to the policy, which replans from the corrected state as the leaders resume tracking.

B. Intervention Detection and Recording

JoyLo+ uses the takeover state to switch control and label corrective experience. The operator decides when recovery is needed and a shared label captures single-arm and coordinated whole-body interventions across all body parts. At each step, we record the observation, robot-interface command, and policy’s action. The command serves as the action target, including human inputs and takeover transitions. We also retain the policy’s proposed action. We separately label original demonstrations, active interventions, 15 frames preceding takeover as pre-intervention context, and autonomous execution to compare rollout-segment selection (Section IV-E). Active intervention frames ($s_t = 2$) form the corrective subset for aggregation (Section III-C).

C. Iterative Interactive Imitation Learning

a) Correction aggregation: Starting from a demonstration dataset $\mathcal{D}_0$, round r deploys the previous policy and collects a labeled rollout dataset $\mathcal{R}_r$ through JoyLo+. Under HG-Dagger filtering [12], the training dataset becomes

$$\mathcal{D}_r = \mathcal{D}_{r-1} \cup \{(\mathbf{x}_t, \mathbf{a}_t^{\text{exec}}) \in \mathcal{R}_r : s_t = 2\}. \quad (1)$$

Thus, original demonstrations and active corrections from all rounds are retained. Pre-intervention and on-policy frames are excluded to concentrate training on human demonstrations and corrective supervision, rather than reproducing autonomous rollout behavior. Other data-selection strategies are compared in Section IV-E. Policy architectures and training settings are specified in Section IV-A.

b) Training from a fixed initialization: Each round trains on $\mathcal{D}_r$ from the same starting weights, rather than continuing from the previous round’s trained policy. For the VLA, this means fine-tuning the same pretrained checkpoint in every round; for diffusion policies, it means restarting from the same weight initialization. The resulting policy is then deployed to collect the next round of corrections.

c) Emergent failure-mode curriculum: Policy updates change the state distribution encountered in subsequent rounds. As earlier failures become less frequent, corrections can shift toward downstream stages such as transport and placement. This curriculum is an emergent effect, not a



Fig. 3. **Evaluation tasks and initialization randomization.** Representative images show initial configurations and successful task outcomes. The eight tasks span 29 annotated subtasks (2–6 per task). Across the four real-world R1 Pro tasks, task executions average approximately 83 seconds, with book storage and breakfast preparation averaging close to two minutes each. Yellow mats indicate the R1 Pro base initialization regions as well as where some task-relevant objects are randomized before each episode.

prescribed assignment of subtasks to rounds. Fixed-policy collection can also reach downstream states through human recovery, but does not update the autonomous policy between collection rounds. Section IV-C evaluates this progression.

IV. EXPERIMENTS

Our primary application is real-world whole-body mobile manipulation on R1 Pro. Complementary Franka tasks examine JoyLo+’s single-arm configuration and precision-oriented correction, while simulation provides controlled comparisons of correction collection and learning algorithm choices. We ask: **Q1:** Do targeted corrections address the last mile problem by improving task success over additional demonstrations? **Q2:** How does iteration affect policy performance and human effort? **Q3:** How do whole-body corrections engage the arms, torso, and mobile base? **Q4:** How do correction-learning strategies compare in simulation? **Q5:** How do policy improvement and intervention trends vary across operators with different skills?

A. Experimental Setup

Repeated collection, retraining, physical resets, and evaluation make long-horizon mobile-manipulation studies resource-intensive. We therefore concentrate controlled algorithm comparisons in simulation and use real-world experiments to evaluate the selected protocol across diverse household tasks.

a) Tasks and platforms: Evaluation is conducted on eight tasks (with twenty-nine subtasks) across simulated and real-world domains involving three robot embodiments. Figure 3 shows the initial configuration and a successful execution for each task, along with the randomization applied to the task objects and to the robot base pose at the start of every episode. Figure 5 shows the annotated subtask stages and their intervention patterns across correction rounds.

Controlled simulation (two tasks). A1 Pick-and-Place Strawberries and R1 Pro Open Fridge and Pick Up Radio are simulated tasks. They comprise four and three annotated subtasks, with task executions averaging approximately 31 and 30 seconds, respectively. We set up our simulation environments using BEHAVIOR [18, 36]. In simulation, scripted

oracles generate corrective trajectories using cuRobo motion planning [37], enabling efficient multi-round collection and controlled algorithm comparisons. A human operator decides when to intervene and when to resume policy execution, while the scripted oracle generates the corrective motions. Both tasks are designated for single-round versus multi-round collection and correction-learning comparisons, including residual learning, BC (all data), SIRIUS, and HG-Dagger. These studies isolate collection and learning choices under controlled conditions.

Fixed-base manipulation (two tasks). Franka peg insertion and mug hanging demonstrate JoyLo+ in a single-arm configuration and evaluate iterative correction on precision-oriented, contact-rich tasks, complementing the whole-body R1 Pro experiments. Each task comprises two annotated subtasks, with task executions averaging approximately 24 seconds for peg insertion and 23 seconds for mug hanging.

Whole-body mobile manipulation (four tasks). R1 Pro performs four long-horizon real-world BEHAVIOR tasks [18]: Prepare Make-Ahead Breakfast Bowls, Boxing Books up for Storage, Make Microwave Popcorn, and Turning on Radio, comprising six, five, four, and three annotated subtasks, respectively. These tasks require sequential object interactions and coordinated mobile manipulation. Task executions average approximately 83 seconds across these tasks, with breakfast preparation and book storage averaging about two minutes each.

b) Evaluation protocol: During supervised collection, autonomous completion denotes episodes finished without human takeover. A subtask is counted as intervention-free when no takeover occurs within its annotated interval. These collection metrics are distinct from success in separate autonomous evaluations. All real-world evaluations use 25 rollouts per condition. Simulation evaluations are averaged across five evaluation seeds, with 50 rollouts per seed.

c) Training: A1 and Franka use diffusion policies (DP) [1] from scratch (UNet backbone, DDIM, AdamW, batch size 64, cosine LR). R1 Pro uses a $\pi_{0.5}$ -based VLA [8]. Each round trains on accumulated demonstrations and active corrections, restarting from the same weight initialization for diffusion policies or the same pretrained checkpoint for the

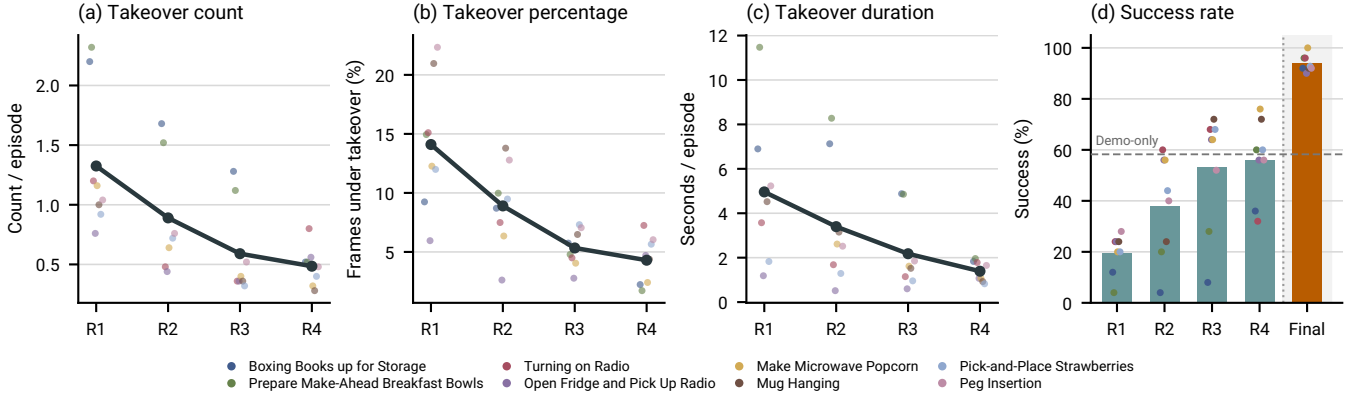


Fig. 4. **Intervention burden and task completion across correction rounds.** In (a–c), the dark line shows the equal-weight mean across eight tasks; (b) measures the fraction of execution under intervention, and (c) measures time under intervention per episode. In (d), rounds 1–4 show takeover-free collection episodes. The final bar and dashed demo-only baseline (58.3%) compare autonomous evaluation success under matched training-trajectory budgets.

VLA. We train all $\pi_{0.5}$ policies for 40k steps and DP for 200k steps before evaluation.

B. Q1: Targeted Corrections Improve Task Performance

TABLE II: Autonomous full-task success before and after additional data collection. Base policies receive 100 additional demonstrations or 100 corrective trajectories through four HG-Dagger rounds. Simulation rates are means across five evaluation seeds.

Task	Base	100 Demos	100 Corr. (Ours)
<i>Simulation</i>			
Pick-and-Place Strawberries	53.2%	55.6%	92.8%
Open Fridge and Pick Up Radio	39.6%	30.4%	90.0%
<i>Real world (Franka, Diffusion Policy)</i>			
Mug Hanging	6/25 (24%)	17/25 (68%)	23/25 (92%)
Peg Insertion	7/25 (28%)	12/25 (48%)	23/25 (92%)
<i>Real world (R1 Pro, VLA)</i>			
Prepare Make-Ahead Breakfast Bowls	1/25 (4%)	19/25 (76%)	24/25 (96%)
Boxing Books up for Storage	3/25 (12%)	13/25 (52%)	23/25 (92%)
Make Microwave Popcorn	5/25 (20%)	20/25 (80%)	25/25 (100%)
Turning on Radio	6/25 (24%)	14/25 (56%)	24/25 (96%)

Table II compares three conditions: the base policy, training with 100 additional expert demonstrations, and training with 100 corrective trajectories collected over four HG-Dagger rounds. Both augmented conditions start from the same base demonstration set. This comparison evaluates full-task reliability when the same additional trajectory budget is allocated to demonstrations or on-policy corrections.

Additional demonstrations improve base-policy success by 32.7 percentage points on average across the eight tasks, while four-round HG-Dagger improves it by 68.3 points, yielding a 35.6-point advantage over collecting 100 additional demonstrations. Four-round HG-Dagger outperforms both baselines on every task, achieving at least 90% success throughout. On the four long-horizon real-world BEHAVIOR tasks [18], corrections provide an average advantage of 30 percentage points over additional demonstrations. These results show that, with the same training-trajectory budget, targeting supervision at deployment-time failures more effectively closes the last-mile reliability gap in whole-body mobile manipulation.

TABLE III: Correction-learning comparison in simulation. Entries report mean $\pm$ sample standard deviation of success rates (%) across five evaluation seeds. Parentheses indicate cumulative corrective trajectories. The entry with the best mean for each task is bold.

Setting	Residual	BC (all data)	SIRIUS	HG-Dagger
<i>Open Fridge and Pick Up Radio (VLA)</i>				
Single round (100)	21.6 $\pm$ 17.5	79.2 $\pm$ 13.2	73.2 $\pm$ 12.5	80.0 $\pm$ 14.8
Round 1 (25)	16.0 $\pm$ 2.0	70.8 $\pm$ 4.6	54.8 $\pm$ 20.9	51.6 $\pm$ 12.3
Round 2 (50)	24.0 $\pm$ 5.1	86.4 $\pm$ 3.8	68.4 $\pm$ 13.1	72.4 $\pm$ 6.2
Round 3 (75)	30.0 $\pm$ 12.2	87.2 $\pm$ 6.6	70.4 $\pm$ 16.9	82.0 $\pm$ 13.6
Round 4 (100)	19.6 $\pm$ 6.2	88.0 $\pm$ 3.7	71.6 $\pm$ 7.9	90.0 $\pm$ 2.4
<i>Pick-and-Place Strawberries (Diffusion Policy)</i>				
Single round (100)	45.2 $\pm$ 2.7	14.8 $\pm$ 2.3	76.0 $\pm$ 2.0	89.2 $\pm$ 3.0
Round 1 (25)	49.2 $\pm$ 4.1	0.4 $\pm$ 0.9	70.8 $\pm$ 8.9	75.6 $\pm$ 6.5
Round 2 (50)	45.6 $\pm$ 1.7	1.2 $\pm$ 1.1	78.8 $\pm$ 4.4	88.4 $\pm$ 3.8
Round 3 (75)	40.0 $\pm$ 7.6	4.4 $\pm$ 2.6	37.6 $\pm$ 6.8	88.4 $\pm$ 4.8
Round 4 (100)	46.8 $\pm$ 9.7	10.8 $\pm$ 5.0	78.8 $\pm$ 5.8	92.8 $\pm$ 2.7

C. Q2: Iteration Improves Performance and Reduces Intervention Burden

a) *Iteration improves success under matched collection budgets:* Table III compares four iterative rounds of 25 episodes with a single batch of 100 episodes collected against the fixed base policy. With HG-Dagger filtering, iterative collection improves success by 3.6 percentage points on Pick-and-Place Strawberries and 10.0 points on Open Fridge and Pick Up Radio. Updating the policy between rounds changes the visited state distribution and allows later corrections to address failures of the updated policy.

b) *Active intervention burden decreases across rounds:* Figure 4 shows a consistent decline in task-averaged intervention burden across the eight tasks. From Round 1 to Round 4, the mean number of takeovers per episode decreases by 63.4%, the mean fraction of frames under intervention decreases by 69.5%, and the mean total takeover duration per episode decreases by 72.0%. Together, these measures indicate that takeovers become less frequent, occupy a smaller fraction of execution, and require less time under intervention per episode. All three task-averaged measures decline in every round, while the final policies achieve at least 90% autonomous success on every task.

c) *Remaining bottlenecks are task dependent:* Although average intervention burden decreases, the remaining correc-

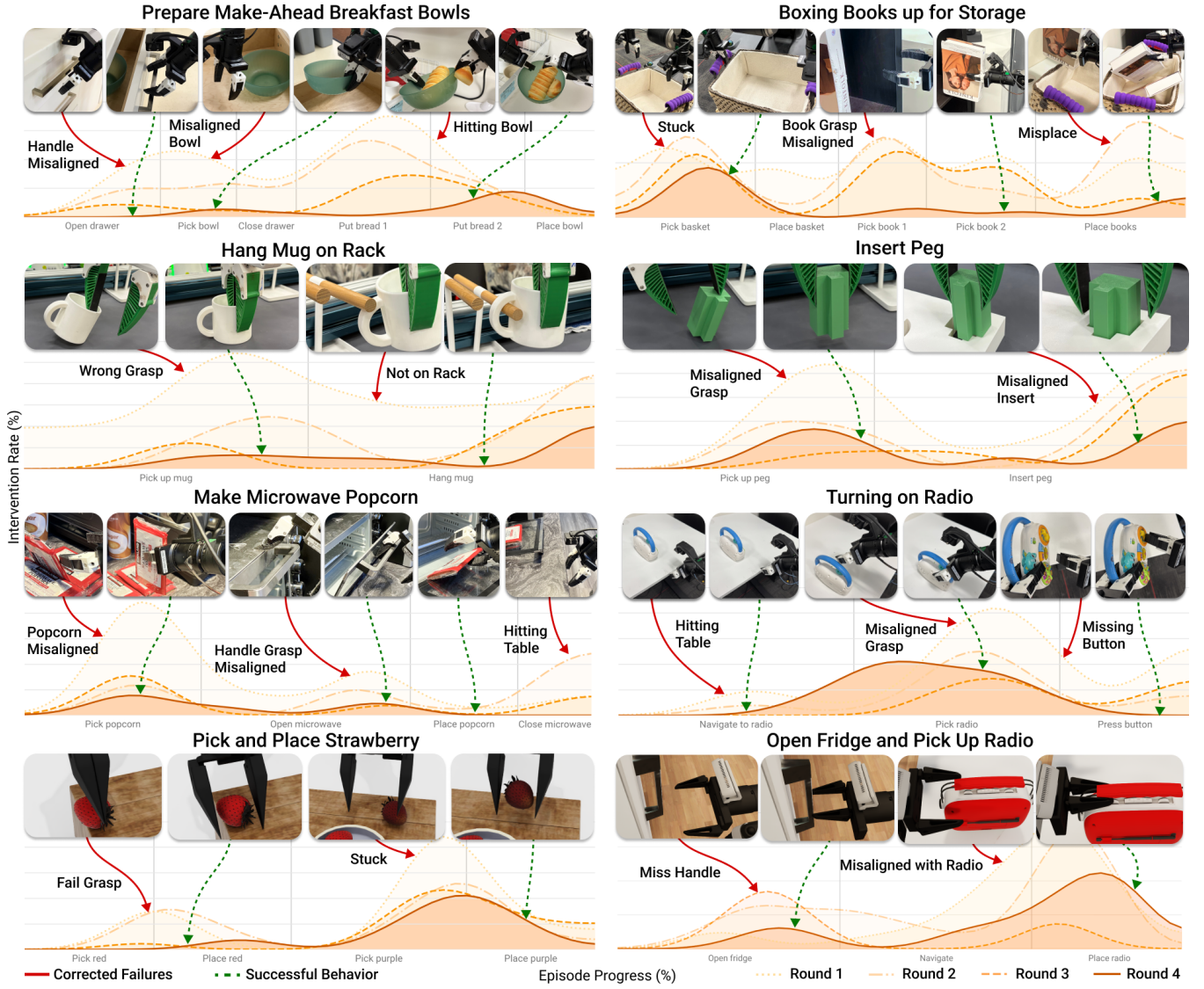


Fig. 5. **Task-stage intervention patterns across correction rounds.** Orange curves show intervention rate over normalized episode progress. The color and line style distinguish Rounds 1–4. Each y-axis tick interval represents 10 percentage points. Red annotations indicate failures and green annotations indicate successful behaviors.

tions concentrate around task-specific difficulties. On Books, Figure 5 shows that intervention around book pickup decreases substantially by Round 4, while a prominent peak remains around basket pickup. On Breakfast Bowls, intervention during drawer opening and bread placement diminishes, while late-stage intervention remains around bowl placement. These two long-horizon BEHAVIOR tasks [18] illustrate how stage-level patterns identify persistent bottlenecks that may benefit from further corrective supervision.

D. Q3: Corrections Span the Whole Body

We analyze how operators use JoyLo+ during interventions on four long-horizon real-world tasks. To characterize whole-body involvement, we identify sustained body-part activity within each takeover segment and normalize the resulting activation counts by task. Multiple body parts can contribute within the same segment.

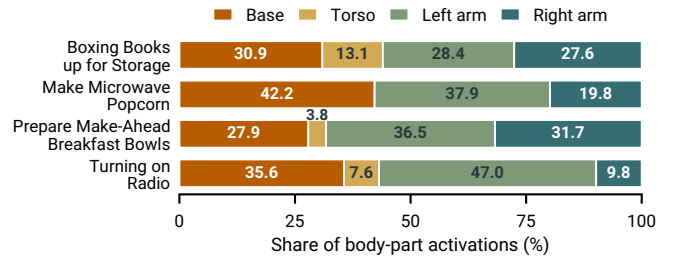


Fig. 6. **Body-part activation shares during JoyLo+ corrections.** Bars pool four 25-episode rounds; each part receives one vote per takeover segment after three consecutive active frames. Thresholds are 0.003 for absolute base delta-position commands and 0.01/0.02/0.75 for torso/arm/gripper action-state deltas, with grippers assigned to their respective arms.

Figure 6 shows that operators use JoyLo+ to correct more than arm motion: base activations account for an average of 34.2% of body-part votes across all four tasks, while

torso activations contribute an average of 8.2% in the three tasks where they occur. Arm correction varies by task, with comparable left and right arm shares in Books and Breakfast, but larger left-arm shares in Radio and Popcorn. These patterns highlight the value of an interface that lets operators engage different combinations of the base, torso, and arms as the task requires.

E. Q4: Correction-Learning Algorithm Comparison

Table III compares HG-Dagger filtering with the following alternatives. Each method collects corrections with its own policy and trains on its accumulated data across rounds.

1) *Residual Policy*: The residual policy takes images, proprioception, and the base policy’s action chunk prediction as input and predicts the residual action between the base policy’s output and actions used for correction. The base policy remains frozen across rounds while the residual model is trained on accumulated corrections. The human demonstrations receive a zero residual target to preserve desired behavior. Since the number of correction timesteps is much smaller than the number of human demonstration or on-policy timesteps, the correction data is upsampled to 50% of the training dataset. The architecture used is a simple MLP with a ResNet [38] backbone.

2) *BC (all data)*: All collected data, including potentially suboptimal on-policy data, is used for training the policy.

3) *SIRIUS*: Following the original SIRIUS sampling scheme [14], intervention frames receive 50% of the sampling probability, demonstrations receive up to 50%, and any remaining probability is assigned to on-policy data.

Four-round HG-Dagger achieves the highest final mean success on both simulation tasks. BC (all data) leads at intermediate rounds and finishes close to HG-Dagger on Open Fridge, but performs poorly on Strawberries throughout collection, consistent with learning suboptimal behavior from on-policy data. SIRIUS improves steadily across rounds on Open Fridge, while its Strawberries performance drops sharply at Round 3 before recovering. Residual policy shows little final benefit from iteration relative to single-round collection on either task. This may reflect the difficulty of learning the large residuals required for some corrections.

F. Q5: Operator Study

We examine policy improvement and intervention burden across three teleoperators with varying levels of experience on Franka Mug Hanging. Each operator collects 25 episodes per round over three correction rounds, totaling 225 episodes. Figure 7 summarizes cumulative intervention burden and takeover-free collection across their respective training histories.

Cumulative intervention share decreases for all three operators, from 16.1% to 11.8%, 34.6% to 19.9%, and 16.4% to 14.7%. By Round 3, takeover-free episodes account for 40–55% of each operator’s cumulative collection. Together, these results suggest that I3L enables iterative policy refinement with declining intervention burden across users with varied teleoperation experience.

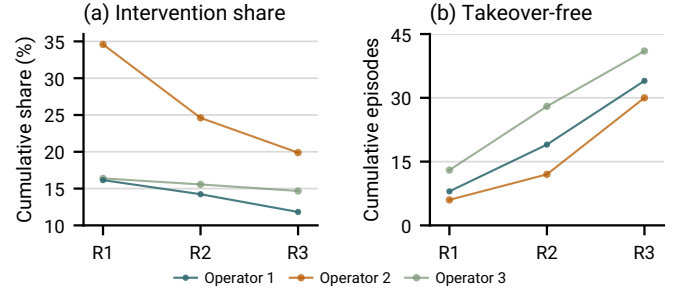


Fig. 7. **Operator study on Franka mug hanging.** (a) Cumulative fraction of timesteps under human control. (b) Cumulative takeover-free episode counts. Counts describe retained collection episodes and do not establish task success.

V. CONCLUSION AND LIMITATIONS

We presented I3L, an integrated hardware and learning system that couples JoyLo+ whole-body takeover with iterative corrective learning for long-horizon mobile manipulation. Across eight tasks, I3L achieves at least 90% autonomous success, outperforming demonstration-only training by an average of 35.6 percentage points under matched training-trajectory budgets. The mean duration of takeover per episode decreases by 72% from the first to the fourth correction round and controlled comparisons further show that both the collection schedule and learning strategy matter, with four-round HG-Dagger achieving the highest final success on both simulation tasks.

Our evaluation spans eight tasks, with controlled algorithm comparisons in simulation, real-world validation of the selected protocol, and a three-operator study. Broader task and operator coverage remains an important direction, given the resource demands of multi-round whole-body experiments. Each round currently restarts training from a common initialization or pretrained checkpoint; incremental updates could reduce retraining cost while preserving performance. We measure active takeover burden, while operators continue to supervise execution. Future work will quantify total operator time and explore uncertainty- and failure-aware assistance.

REFERENCES

- [1] C. Chi, S. Feng, Y. Du, Z. Xu, E. Cousineau, B. Burchfiel, and S. Song, “Diffusion policy: Visuomotor policy learning via action diffusion,” in *Proceedings of Robotics: Science and Systems (RSS)*, 2023.
- [2] T. Z. Zhao, V. Kumar, S. Levine, and C. Finn, “Learning fine-grained bimanual manipulation with low-cost hardware,” in *Proceedings of Robotics: Science and Systems (RSS)*, 2023.
- [3] B. Zitkovich, T. Yu, S. Xu, P. Xu, *et al.*, “RT-2: Vision-language-action models transfer web knowledge to robotic control,” in *Proceedings of the 7th Conference on Robot Learning (CoRL)*, ser. Proceedings of Machine Learning Research, vol. 229, 2023, pp. 2165–2183.
- [4] Open X-Embodiment Collaboration, A. O’Neill, *et al.*, “Open X-Embodiment: Robotic learning datasets and RT-X models,” in *IEEE International Conference on Robotics and Automation (ICRA)*, 2024, pp. 6892–6903.
- [5] M. J. Kim, K. Pertsch, S. Karamcheti, T. Xiao, A. Balakrishna, S. Nair, R. Rafailov, E. Foster, P. R. Sanketi, Q. Vuong, T. Kollar, B. Burchfiel, R. Tedrake, D. Sadigh, S. Levine, P. Liang, and C. Finn, “OpenVLA: An open-source vision-language-action model,” in *Proceedings of the 8th Conference on Robot Learning (CoRL)*, ser. Proceedings of Machine Learning Research, vol. 270, 2025, pp. 2679–2713.
- [6] Octo Model Team, D. Ghosh, H. Walke, K. Pertsch, K. Black, O. Mees, S. Dasari, J. Hejna, T. Kreiman, C. Xu, J. Luo, Y. L. Tan, L. Y. Chen, P. Sanketi, Q. Vuong, T. Xiao, D. Sadigh, C. Finn,

- and S. Levine, "Octo: An open-source generalist robot policy," in *Proceedings of Robotics: Science and Systems (RSS)*, 2024.
- [7] K. Black, N. Brown, D. Driess, A. Esmail, M. Equi, C. Finn, *et al.*, " π_0 : A vision-language-action flow model for general robot control," in *Proceedings of Robotics: Science and Systems (RSS)*, 2025.
 - [8] Physical Intelligence, K. Black, N. Brown, J. Darpinian, K. Dhabalia, D. Driess, A. Esmail, M. Equi, C. Finn, N. Fusai, M. Y. Galliker, D. Ghosh, L. Groom, K. Hausman, B. Ichter, S. Jakubczak, T. Jones, L. Ke, D. LeBlanc, S. Levine, A. Li-Bell, M. Mothukuri, S. Nair, K. Pertsch, A. Z. Ren, L. X. Shi, L. Smith, J. T. Springenberg, K. Stachowicz, J. Tanner, Q. Vuong, H. Walke, A. Walling, H. Wang, L. Yu, and U. Zhilinsky, " $\pi_{0.5}$: a vision-language-action model with open-world generalization," *arXiv preprint arXiv:2504.16054*, 2025.
 - [9] NVIDIA, J. Bjorck, F. Castañeda, N. Cherniadev, X. Da, R. Ding, L. Fan, Y. Fang, D. Fox, F. Hu, S. Huang, J. Jang, Z. Jiang, J. Kautz, K. Kundalia, L. Lao, Z. Li, Z. Lin, K. Lin, G. Liu, E. Llontop, L. Magne, A. Mandlekar, A. Narayan, S. Nasiriany, S. Reed, Y. L. Tan, G. Wang, Z. Wang, Q. Wang, J. Xiang, Y. Xie, Y. Xu, Z. Xu, S. Ye, Z. Yu, A. Zhang, H. Zhang, Y. Zhao, R. Zheng, and Y. Zhu, "GR00T N1: An open foundation model for generalist humanoid robots," *arXiv preprint arXiv:2503.14734*, 2025.
 - [10] S. Ross, G. J. Gordon, and J. A. Bagnell, "A reduction of imitation learning and structured prediction to no-regret online learning," in *Proceedings of the Fourteenth International Conference on Artificial Intelligence and Statistics (AISTATS)*, ser. Proceedings of Machine Learning Research, vol. 15, 2011, pp. 627–635.
 - [11] M. Laskey, J. Lee, R. Fox, A. Dragan, and K. Goldberg, "DART: Noise injection for robust imitation learning," in *Proceedings of the 1st Annual Conference on Robot Learning (CoRL)*, ser. Proceedings of Machine Learning Research, vol. 78, 2017, pp. 143–156.
 - [12] M. Kelly, C. Sidrane, K. Driggs-Campbell, and M. J. Kochenderfer, "HG-DAGger: Interactive imitation learning with human experts," in *IEEE International Conference on Robotics and Automation (ICRA)*, 2019, pp. 8077–8083.
 - [13] A. Mandlekar, D. Xu, R. Martín-Martín, Y. Zhu, L. Fei-Fei, and S. Savarese, "Human-in-the-loop imitation learning using remote teleoperation," *arXiv preprint arXiv:2012.06733*, 2020.
 - [14] H. Liu, S. Nasiriany, L. Zhang, Z. Bao, and Y. Zhu, "Robot learning on the job: Human-in-the-loop autonomy and learning during deployment," in *Proceedings of Robotics: Science and Systems (RSS)*, 2023.
 - [15] Y. Jiang, C. Wang, R. Zhang, J. Wu, and L. Fei-Fei, "TRANSIC: Sim-to-real policy transfer by learning from online correction," in *Proceedings of the 8th Conference on Robot Learning (CoRL)*, ser. Proceedings of Machine Learning Research, vol. 270, 2025, pp. 1691–1729.
 - [16] P. Wu, Y. Shentu, Q. Liao, D. Jin, M. Guo, K. Sreenath, X. Lin, and P. Abbeel, "RoboCopilot: Human-in-the-loop interactive imitation learning for robot manipulation," *arXiv preprint arXiv:2503.07771*, 2025.
 - [17] X. Xu, Y. Hou, Z. Liu, and S. Song, "Compliant residual DAGger: Improving real-world contact-rich manipulation with human corrections," in *Advances in Neural Information Processing Systems (NeurIPS)*, vol. 38, 2025. [Online]. Available: <https://doi.org/10.52202/085713-4660>
 - [18] C. Li, R. Zhang, J. Wong, C. Gokmen, S. Srivastava, R. Martín-Martín, C. Wang, G. Levine, W. Ai, B. Martinez, H. Yin, M. Lingelbach, M. Hwang, A. Hiranaka, S. Garlanka, A. Aydin, S. Lee, J. Sun, M. Anvari, M. Sharma, D. Bansal, S. Hunter, K.-Y. Kim, A. Lou, C. R. Matthews, I. Villa-Renteria, J. H. Tang, C. Tang, F. Xia, Y. Li, S. Savarese, H. Gweon, C. K. Liu, J. Wu, and L. Fei-Fei, "BEHAVIOR-1K: A human-centered, embodied AI benchmark with 1,000 everyday activities and realistic simulation," *arXiv preprint arXiv:2403.09227*, 2024. [Online]. Available: <https://arxiv.org/abs/2403.09227>
 - [19] P. Wu, Y. Shentu, Z. Yi, X. Lin, and P. Abbeel, "GELLO: A general, low-cost, and intuitive teleoperation framework for robot manipulators," in *IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*, 2024, pp. 12 156–12 163.
 - [20] Y. Jiang, R. Zhang, J. Wong, C. Wang, Y. Ze, H. Yin, C. Gokmen, S. Song, J. Wu, and L. Fei-Fei, "BEHAVIOR robot suite: Streamlining real-world whole-body manipulation for everyday household activities," in *Proceedings of the 9th Conference on Robot Learning (CoRL)*, ser. Proceedings of Machine Learning Research, vol. 305. PMLR, 2025, pp. 1246–1281. [Online]. Available: <https://proceedings.mlr.press/v305/jiang25b.html>
 - [21] C. Wang, H. Shi, W. Wang, R. Zhang, L. Fei-Fei, and C. K. Liu, "DexCap: Scalable and portable mocap data collection system for dexterous manipulation," in *Proceedings of Robotics: Science and Systems (RSS)*, 2024.
 - [22] S. Dass, W. Ai, Y. Jiang, S. Singh, J. Hu, R. Zhang, P. Stone, B. Abbatematteo, and R. Martín-Martín, "TeleMoMa: A modular and versatile teleoperation system for mobile manipulation," *arXiv preprint arXiv:2403.07869*, 2024.
 - [23] Z. Fu, T. Z. Zhao, and C. Finn, "Mobile ALOHA: Learning bimanual mobile manipulation using low-cost whole-body teleoperation," in *Proceedings of the 8th Conference on Robot Learning (CoRL)*, ser. Proceedings of Machine Learning Research, vol. 270, 2025, pp. 4066–4083.
 - [24] Z. Xu, Y. Zhao, K. Wu, N. Liu, J. Ji, Z. Che, C. H. Liu, and J. Tang, "HACTS: a human-as-copilot teleoperation system for robot learning," in *IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*, 2025, pp. 15 475–15 481.
 - [25] R. Yan, Z. Gongye, L. Paulsen, X. Cheng, and X. Wang, "Human-robot copilot for data-efficient imitation learning," *arXiv preprint arXiv:2604.03613*, 2026.
 - [26] Y. Han, Z. Chen, Y. Zhao, C. Xu, Y. Shao, Y. Peng, Y. Mu, and W. Lian, "DexHIL: A human-in-the-loop framework for vision-language-action model post-training in dexterous manipulation," *arXiv preprint arXiv:2603.09121*, 2026.
 - [27] E. Jang, A. Irpan, M. Khansari, D. Kappler, F. Ebert, C. Lynch, S. Levine, and C. Finn, "BC-Z: Zero-shot task generalization with robotic imitation learning," in *Proceedings of the 5th Conference on Robot Learning (CoRL)*, ser. Proceedings of Machine Learning Research, vol. 164, 2022, pp. 991–1002.
 - [28] J. Zhang and K. Cho, "Query-efficient imitation learning for end-to-end simulated driving," in *Proceedings of the AAAI Conference on Artificial Intelligence (AAAI)*, 2017, pp. 2891–2897.
 - [29] R. Hoque, A. Balakrishna, E. Novoseller, A. Wilcox, D. S. Brown, and K. Goldberg, "ThriftyDAGger: Budget-aware novelty and risk gating for interactive imitation learning," in *Proceedings of the 5th Conference on Robot Learning (CoRL)*, ser. Proceedings of Machine Learning Research, vol. 164. PMLR, 2022, pp. 598–608.
 - [30] E. Welte, Y. Shi, R. Wolf, M. Gilles, and R. Rayyes, "FlowCorrect: Efficient interactive correction of generative flow policies for robotic manipulation," *arXiv preprint arXiv:2602.22056*, 2026.
 - [31] Physical Intelligence, " $\pi_{0.6}^*$: a VLA that learns from experience," 2025. [Online]. Available: <https://www.pi.website/download/pistar06.pdf>
 - [32] Z. Fu, Q. Zhao, Q. Wu, G. Wetzstein, and C. Finn, "HumanPlus: Humanoid shadowing and imitation from humans," in *Proceedings of the 8th Conference on Robot Learning (CoRL)*, ser. Proceedings of Machine Learning Research, vol. 270, 2025, pp. 2828–2844.
 - [33] J. Xu, Q. Li, Y. Shen, Y. Ren, T. Davies, S. He, Z. Wang, Y. Yang, R. Cheng, and H. Dong, "Scaling bimanual household manipulation from 1,500 hours of demonstrations to on-policy corrections," *arXiv preprint arXiv:2609.03591*, 2026.
 - [34] ALOHA 2 Team, "ALOHA 2: An enhanced low-cost hardware for bimanual teleoperation," *arXiv preprint arXiv:2405.02292*, 2024.
 - [35] L. Liang, J. Xu, X. Qi, Y. Cai, H. Ding, L. Wang, Z. Sun, J.-H. Chow, M. Yang, and M. Cutkosky, "Universal manipulation exoskeleton: Learning compliant whole-body policies with real-time torque feedback," *arXiv preprint arXiv:2606.14218*, 2026.
 - [36] Y. Ge, Y. Tang, J. Xu, C. Gokmen, C. Li, W. Ai, B. J. Martinez, A. Aydin, M. Anvari, A. K. Chakravarthy, H.-X. Yu, J. Wong, S. Srivastava, S. Lee, S. Zha, L. Itti, Y. Li, R. Martín-Martín, M. Liu, P. Zhang, R. Zhang, L. Fei-Fei, and J. Wu, "BEHAVIOR Vision Suite: Customizable dataset generation via simulation," in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2024, pp. 22 401–22 412.
 - [37] B. Sundaralingam, S. K. S. Hari, A. Fishman, C. Garrett, K. Van Wyk, V. Blukis, A. Millane, H. Oleynikova, A. Handa, F. Ramos, N. Ratliff, and D. Fox, "cuRobo: Parallelized collision-free minimum-jerk robot motion generation," *arXiv preprint arXiv:2310.17274*, 2023.
 - [38] K. He, X. Zhang, S. Ren, and J. Sun, "Deep residual learning for image recognition," in *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, 2016, pp. 770–778.